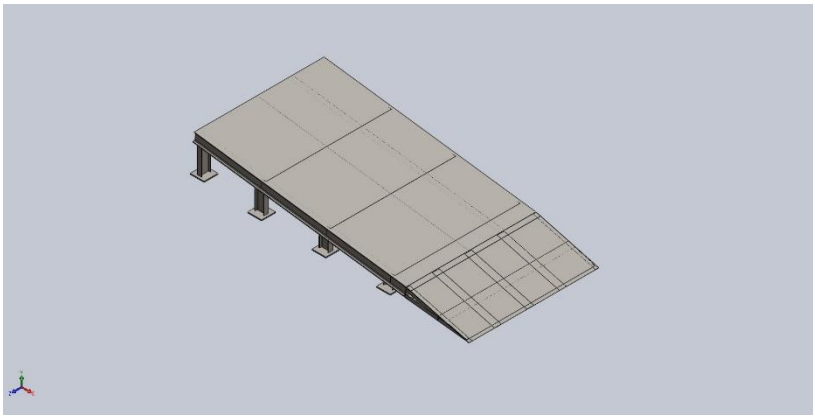


Simulation of 0403Assem1

Date: 15 APRIL 2025
Designer: BHARATI SABALE
Study name: Static 2
Analysis type: Static

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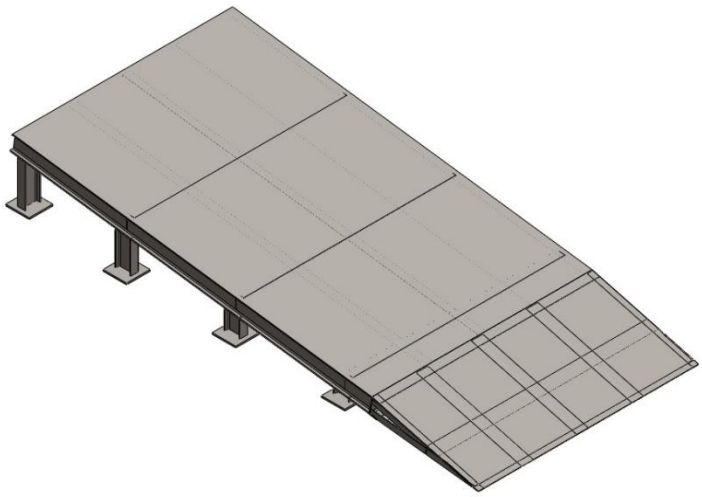
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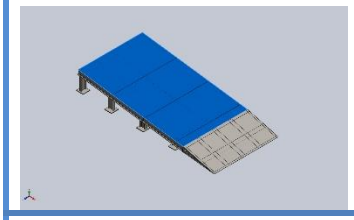
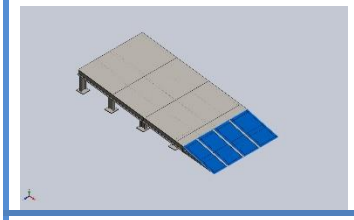
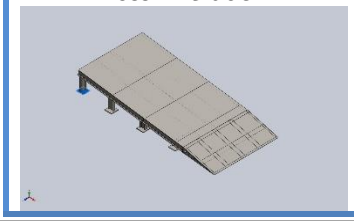
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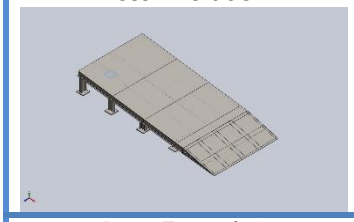
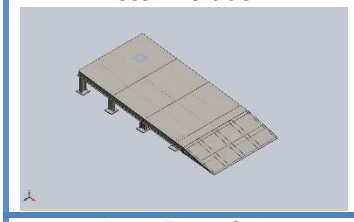
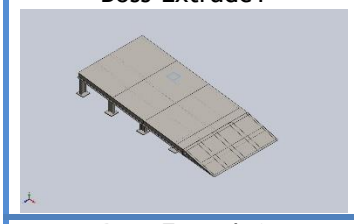
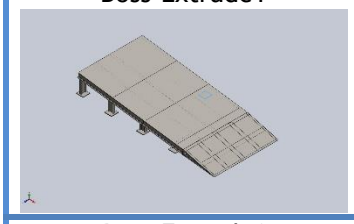
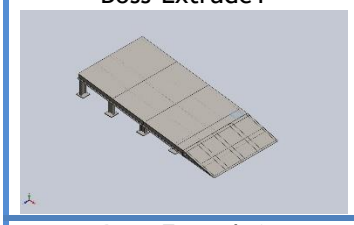
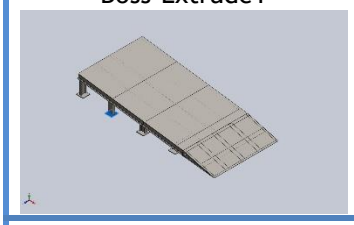
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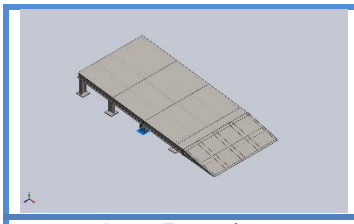
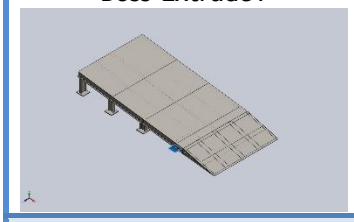


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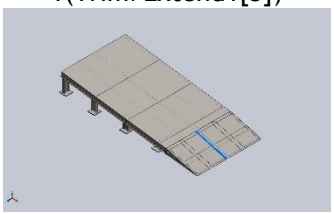
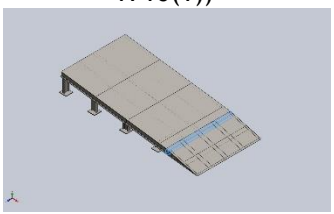
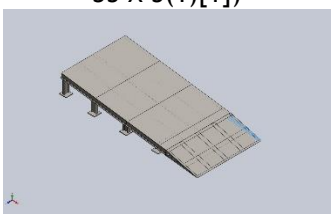
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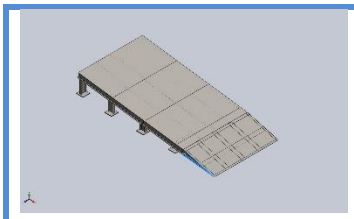
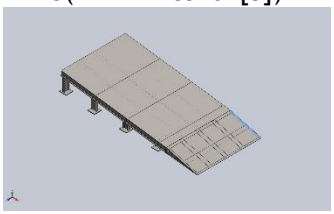
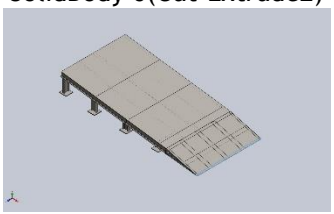
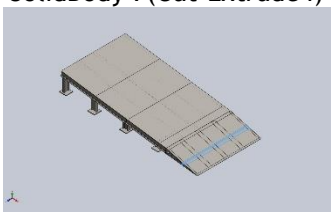
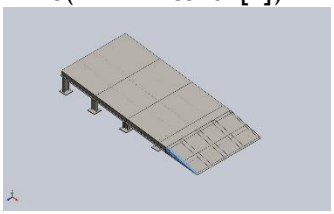
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Boss-Extrude1 	Solid Body	**Mass:126.66 kg** **Volume:0.016135 m^3** **Density:7,850 kg/m^3** **Weight:1,241.27 N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403CHEQURED PLATE.SLDPRT** Oct 19 18:40:39 2025
Boss-Extrude1 	Solid Body	**Mass:38.465 kg** **Volume:0.0049 m^3** **Density:7,850 kg/m^3** **Weight:376.957 N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403SS PLATE.SLDPRT** Oct 21 06:22:15 2025
Boss-Extrude1 	Solid Body	**Mass:1.76625 kg** **Volume:0.000225 m^3** **Density:7,850 kg/m^3** **Weight:17.3092 N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403bottom plate.SLDPRT** Oct 19 18:40:39 2025

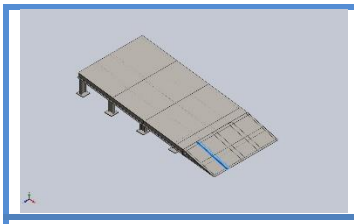
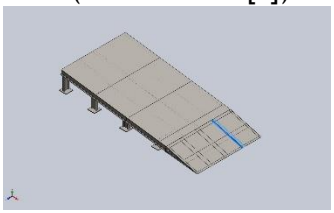
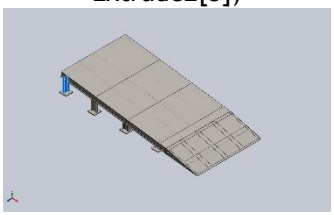
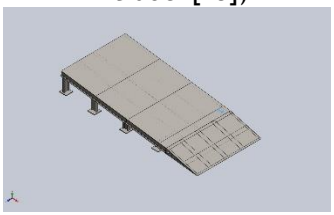
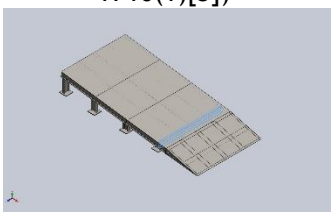
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Boss-Extrude1 	Solid Body	**Mass:1.76625 kg Volume:0.000225 m^3 Density:7,850 kg/m^3 Weight:17.3092 N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403bottom plate.SLDPRT** Oct 19 18:40:39 2025
Boss-Extrude1 	Solid Body	**Mass:1.76625 kg Volume:0.000225 m^3 Density:7,850 kg/m^3 Weight:17.3093 N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403bottom plate.SLDPRT** Oct 19 18:40:39 2025
Boss-Extrude1 	Solid Body	**Mass:1.76625 kg Volume:0.000225 m^3 Density:7,850 kg/m^3 Weight:17.3092 N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403bottom plate.SLDPRT** Oct 19 18:40:39 2025
Boss-Extrude1 	Solid Body	**Mass:1.76625 kg Volume:0.000225 m^3 Density:7,850 kg/m^3 Weight:17.3092 N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403bottom plate.SLDPRT** Oct 19 18:40:39 2025
Boss-Extrude1 	Solid Body	**Mass:1.76625 kg Volume:0.000225 m^3 Density:7,850 kg/m^3 Weight:17.3092 N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403bottom plate.SLDPRT** Oct 19 18:40:39 2025
Boss-Extrude1	Solid Body	**Mass:1.76625 kg Volume:0.000225 m^3 Density:7,850 kg/m^3 Weight:17.3092 N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA**

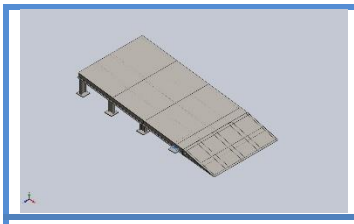
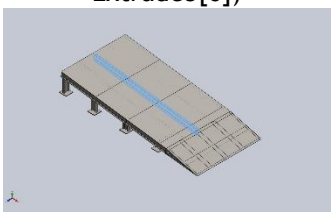
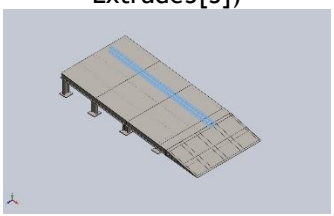
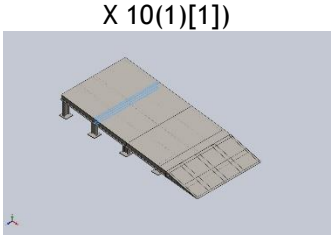
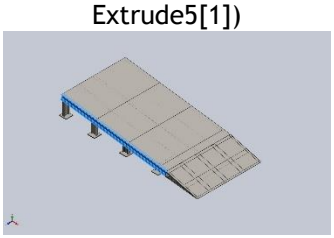
			**ABLOY 04\0403bottom plate.SLDPRT** Oct 19 18:40:39 2025
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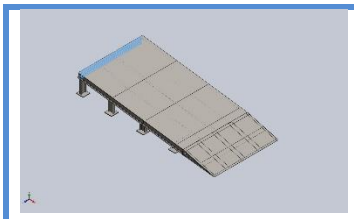
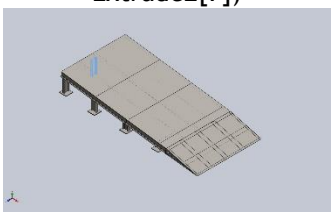
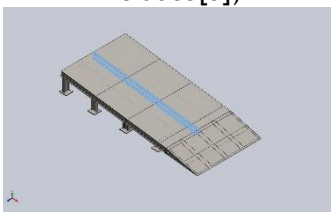
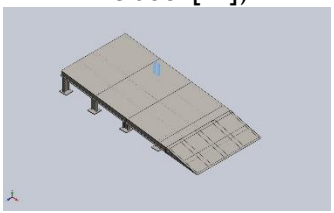
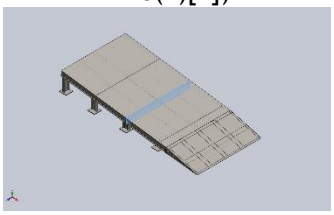
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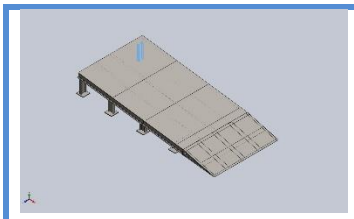
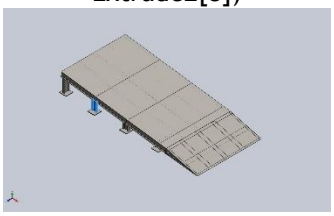
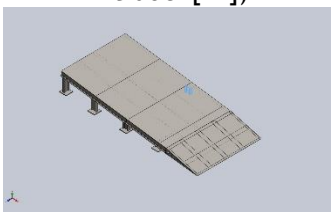
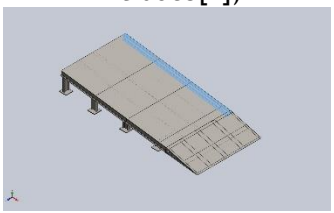
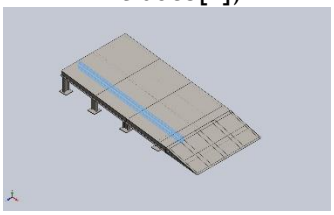
Document Name and Reference	Formulation	Properties	Document Path/Date Modified
SolidBody 1(Trim/Extend1[3]) 	Beam - Uniform C/S	**Section Standard-** **Section Area: 0.000856274m^2** **Length:693.682mm** **Volume:0.000479396m^3** **Mass Density:7,850kg/m^3** **Mass:3.76326kg** **Weight:36.88N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403DOCK RAMP2.SLDPRT** Oct 21 06:22:14 2025
SolidBody 2(C channel 100 X 10(1)) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/c channel** **Section Area: 0.00131018m^2** **Length:1,400mm** **Volume:0.00183425m^3** **Mass Density:7,850kg/m^3** **Mass:14.3988kg** **Weight:141.109N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403DOCK RAMP2.SLDPRT** Oct 21 06:22:14 2025
SolidBody 3(Angle iron 35 X 35 X 5(1)[1]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/angle iron** **Section Area: 0.000319635m^2** **Length:676.965mm** **Volume:0.000201915m^3** **Mass Density:7,850kg/m^3** **Mass:1.58504kg** **Weight:15.5333N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403DOCK RAMP2.SLDPRT** Oct 21 06:22:14 2025
SolidBody 4(Angle iron 35 X 35 X 5(1)[2])	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/angle iron**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA**

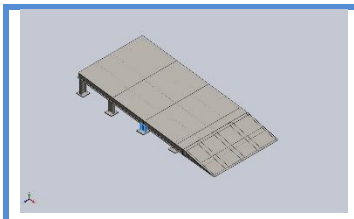
		**Section Area: 0.000319635m^2 Length:668.363mm Volume:0.00020192m^3 Mass Density:7,850kg/m^3 Mass:1.58507kg Weight:15.5337N**	**ABLOY 04\0403DOCK RAMP2.SLDPRT** Oct 21 06:22:14 2025
SolidBody 5(Trim/Extend1[5]) 	Beam - Uniform C/S	**Section Standard- Section Area: 0.000319635m^2 Length:650.834mm Volume:0.000155116m^3 Mass Density:7,850kg/m^3 Mass:1.21766kg Weight:11.9331N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403DOCK RAMP2.SLDPRT** Oct 21 06:22:14 2025
SolidBody 6(Cut-Extrude2) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/angle iron Section Area: 8.65191e-05m^2 Length:1,400mm Volume:0.000121127m^3 Mass Density:7,850kg/m^3 Mass:0.950845kg Weight:9.31828N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403DOCK RAMP2.SLDPRT** Oct 21 06:22:14 2025
SolidBody 7(Cut-Extrude4) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/angle iron Section Area: 0.000715454m^2 Length:1,400mm Volume:0.00100164m^3 Mass Density:7,850kg/m^3 Mass:7.86284kg Weight:77.0558N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403DOCK RAMP2.SLDPRT** Oct 21 06:22:14 2025
SolidBody 8(Trim/Extend1[1]) 	Beam - Uniform C/S	**Section Standard- Section Area: 0.000319635m^2 Length:650.834mm Volume:0.000155116m^3 Mass Density:7,850kg/m^3 Mass:1.21766kg Weight:11.9331N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403DOCK RAMP2.SLDPRT** Oct 21 06:22:14 2025
SolidBody 9(Trim/Extend1[2])	Beam - Uniform C/S	**Section Standard- Section Area: 0.000856274m^2 Length:693.682mm**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA**

		**Volume:0.000479396m^3 Mass Density:7,850kg/m^3 Mass:3.76326kg Weight:36.88N**	**ABLOY 04\0403DOCK RAMP2.SLDPRT** Oct 21 06:22:14 2025
SolidBody 10(Trim/Extend1[4]) 	Beam - Uniform C/S	**Section Standard- Section Area: 0.000856274m^2 Length:693.682mm Volume:0.000479396m^3 Mass Density:7,850kg/m^3 Mass:3.76326kg Weight:36.88N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403DOCK RAMP2.SLDPRT** Oct 21 06:22:14 2025
SolidBody 1(Cut-Extrude2[5]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.00109003m^2 Length:300mm Volume:0.000327009m^3 Mass Density:7,850kg/m^3 Mass:2.56702kg Weight:25.1568N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 2(Cut-Extrude2[13]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.000107189m^2 Length:92.0026mm Volume:2.59644e-05m^3 Mass Density:7,850kg/m^3 Mass:0.20382kg Weight:1.99744N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 3(C channel 100 X 10(1)[3]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/c channel Section Area: 0.00131018m^2 Length:1,350mm Volume:0.00176874m^3 Mass Density:7,850kg/m^3 Mass:13.8846kg Weight:136.069N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 4(Cut-Extrude2[10])	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.000107189m^2**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA**

		**Length:92.0026mm Volume:2.59644e-05m^3 Mass Density:7,850kg/m^3 Mass:0.20382kg Weight:1.99744N**	**ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 5(Cut-Extrude5[6]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.00109003m^2 Length:2,265.04mm Volume:0.0024703m^3 Mass Density:7,850kg/m^3 Mass:19.3919kg Weight:190.04N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 6(Cut-Extrude5[5]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.00109003m^2 Length:2,265.04mm Volume:0.0024703m^3 Mass Density:7,850kg/m^3 Mass:19.3919kg Weight:190.04N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 7(C channel 100 X 10(1)[1]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/c channel Section Area: 0.00131018m^2 Length:1,350mm Volume:0.00176874m^3 Mass Density:7,850kg/m^3 Mass:13.8846kg Weight:136.069N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 8(Cut-Extrude5[1]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.00109003m^2 Length:2,307.92mm Volume:0.00251577m^3 Mass Density:7,850kg/m^3 Mass:19.7488kg Weight:193.539N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 9(Sb beam 100 X 8(1)[4])	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA**

		**Section Area: 0.00109003m^2 Length:1,300mm Volume:0.00141704m^3 Mass Density:7,850kg/m^3 Mass:11.1238kg Weight:109.013N**	**ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 10(Cut-Extrude2[7]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.00109003m^2 Length:300mm Volume:0.000327009m^3 Mass Density:7,850kg/m^3 Mass:2.56702kg Weight:25.1568N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 11(Cut-Extrude5[3]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.00109003m^2 Length:2,282.6mm Volume:0.00248845m^3 Mass Density:7,850kg/m^3 Mass:19.5343kg Weight:191.436N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 12(Cut-Extrude2[11]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.00109003m^2 Length:207.94mm Volume:0.000226661m^3 Mass Density:7,850kg/m^3 Mass:1.77929kg Weight:17.437N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 13(C channel 100 X 10(1)[2]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/c channel Section Area: 0.00131018m^2 Length:1,350mm Volume:0.00176874m^3 Mass Density:7,850kg/m^3 Mass:13.8846kg Weight:136.069N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 14(Cut-Extrude2[6])	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK**

		**Section Area: 0.00109003m^2 Length:300mm Volume:0.000327009m^3 Mass Density:7,850kg/m^3 Mass:2.56702kg Weight:25.1568N**	**RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 15(Cut-Extrude2[8]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.00109003m^2 Length:207.94mm Volume:0.000226661m^3 Mass Density:7,850kg/m^3 Mass:1.77929kg Weight:17.437N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 16(Cut-Extrude2[12]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.00109003m^2 Length:115.88mm Volume:0.000126313m^3 Mass Density:7,850kg/m^3 Mass:0.991555kg Weight:9.71723N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 17(Cut-Extrude5[2]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.00109003m^2 Length:2,307.92mm Volume:0.00251577m^3 Mass Density:7,850kg/m^3 Mass:19.7488kg Weight:193.539N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 18(Cut-Extrude5[4]) 	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam Section Area: 0.00109003m^2 Length:2,265.04mm Volume:0.0024703m^3 Mass Density:7,850kg/m^3 Mass:19.3919kg Weight:190.04N**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025
SolidBody 19(Cut-Extrude2[9])	Beam - Uniform C/S	**Section Standard-weldment profiles/iso/sb beam**	**E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK**

		**Section Area:** **0.00109003m^2** **Length:115.88mm** **Volume:0.000126313m^3** **Mass Density:7,850kg/m^3** **Mass:0.991555kg** **Weight:9.71723N**	**RAMP\DOCK RAMP- ASSA ABLOY 04\0403sissior lift ramp.SLDPRT** Oct 21 14:04:33 2025

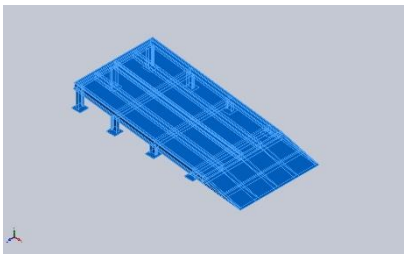
Study Properties

Study name	Static 2
Analysis type	Static
Mesh type	Mixed Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (E:\NIHVA AUTOMATION 2026\AUTOMATION\DOCK RAMP\DOCK RAMP- ASSA ABLOY 04)

Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m^2

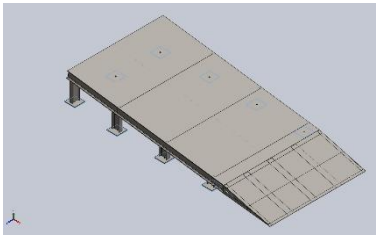
Material Properties

Model Reference	Properties	Components
	**Name:** AISI 1045 Steel, cold drawn **Model type:** Linear Elastic Isotropic **Default failure criterion:** Max von Mises Stress **Yield strength:** 5.3e+08 N/m^2 **Tensile strength:** 6.25e+08 N/m^2 **Elastic modulus:** 2.05e+11 N/m^2 **Poisson's ratio:** 0.29 **Mass density:** 7,850 kg/m^3 **Shear modulus:** 8e+10 N/m^2 **Thermal expansion coefficient:** 1.15e-05 /Kelvin	SolidBody 1(Boss-Extrude1)(0403CHEQURED PLATE-1), SolidBody 1(Trim/Extend1[3])(0403DOCK RAMP2-1), SolidBody 2(C channel 100 X 10(1))(0403DOCK RAMP2-1), SolidBody 3(Angle iron 35 X 35 X 5(1)[1])(0403DOCK RAMP2-1), SolidBody 4(Angle iron 35 X 35 X 5(1)[2])(0403DOCK RAMP2-1), SolidBody 5(Trim/Extend1[5])(0403DOCK RAMP2-1), SolidBody 6(Cut-Extrude2)(0403DOCK RAMP2-1), SolidBody 7(Cut-Extrude4)(0403DOCK RAMP2-1), SolidBody 8(Trim/Extend1[1])(0403DOCK RAMP2-1), SolidBody 9(Trim/Extend1[2])(0403DOCK RAMP2-1), SolidBody 10(Trim/Extend1[4])(0403DOCK RAMP2-1), SolidBody 1(Boss-Extrude1)(0403SS PLATE-1), SolidBody 1(Boss-Extrude1)(0403bottom plate-1), SolidBody 1(Boss-Extrude1)(0403bottom plate-13), SolidBody 1(Boss-Extrude1)(0403bottom plate-15),

		SolidBody 1(Boss-Extrude1)(0403bottom plate-17), SolidBody 1(Boss-Extrude1)(0403bottom plate-19), SolidBody 1(Boss-Extrude1)(0403bottom plate-21), SolidBody 1(Boss-Extrude1)(0403bottom plate-3), SolidBody 1(Boss-Extrude1)(0403bottom plate-5), SolidBody 1(Boss-Extrude1)(0403bottom plate-7), SolidBody 1(Cut-Extrude2[5])(0403sissior lift ramp-1), SolidBody 2(Cut-Extrude2[13])(0403sissior lift ramp-1), SolidBody 3(C channel 100 X 10(1)[3])(0403sissior lift ramp-1), SolidBody 4(Cut-Extrude2[10])(0403sissior lift ramp-1), SolidBody 5(Cut-Extrude5[6])(0403sissior lift ramp-1), SolidBody 6(Cut-Extrude5[5])(0403sissior lift ramp-1), SolidBody 7(C channel 100 X 10(1)[1])(0403sissior lift ramp-1), SolidBody 8(Cut-Extrude5[1])(0403sissior lift ramp-1), SolidBody 9(Sb beam 100 X 8(1)[4])(0403sissior lift ramp-1), SolidBody 10(Cut-Extrude2[7])(0403sissior lift ramp-1), SolidBody 11(Cut-Extrude5[3])(0403sissior lift ramp-1),

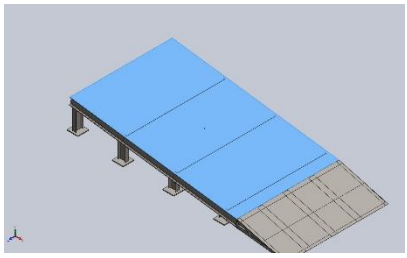
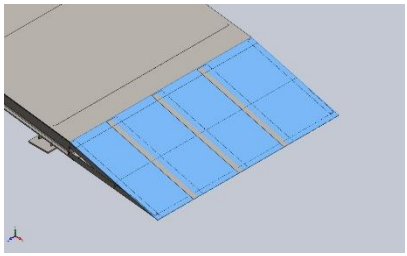
		SolidBody 12(Cut-Extrude2[11])(0403sissior lift ramp-1), SolidBody 13(C channel 100 X 10(1)[2])(0403sissior lift ramp-1), SolidBody 14(Cut-Extrude2[6])(0403sissior lift ramp-1), SolidBody 15(Cut-Extrude2[8])(0403sissior lift ramp-1), SolidBody 16(Cut-Extrude2[12])(0403sissior lift ramp-1), SolidBody 17(Cut-Extrude5[2])(0403sissior lift ramp-1), SolidBody 18(Cut-Extrude5[4])(0403sissior lift ramp-1), SolidBody 19(Cut-Extrude2[9])(0403sissior lift ramp-1)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		**Entities:** 9 face(s) **Type:** Fixed Geometry

Resultant Forces

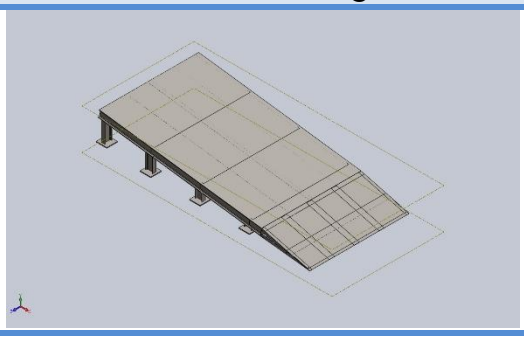
Components	X	Y	Z	Resultant
Reaction force(N)	975.833	4,893.1	0.000304222	4,989.46
Reaction Moment(N.m)	0	0	0	1e-33

Load name	Load Image	Load Details
Force-1		**Entities:** 1 face(s) **Type:** Apply normal force **Value:** 2,500 N
Force-2		**Entities:** 1 face(s) **Type:** Apply normal force **Value:** 2,500 N

Connector Definitions

No Data

Interaction Information

Interaction	Interaction Image	Interaction Properties
Global Interaction		**Type:** Bonded **Components:** 1 component(s) **Options:** Independent mesh

Mesh information

Mesh type	Mixed Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Jacobian check for shell	Off
Maximum element size	59.7697 mm
Minimum element size	59.7697 mm
Mesh Quality	High
Remesh failed parts independently	Off

Mesh information - Details

Total Nodes	19451
Total Elements	9542
Time to complete mesh(hh;mm;ss):	00:00:09
Computer name:	

Sensor Details

No Data

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	975.833	4,893.1	0.000304222	4,989.46

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.671347	1.1651	0.00165558	1.34468

Free body moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

Beam Forces

Beam Name	Joints	Axial(N)	Shear1(N)	Shear2(N)	Moment1(N.m)	Moment2(N.m)	Torque(N.m)
Beam-1(Trim/Extend1[3])	1	-52.2036	-22.677	-37.9599	-13.0097	8.2271	-0.0437733
	2	1.33141	1.56239	0.183235	-0.0236465	0.0517413	0.00830904
	3	-1.33141	-1.56239	-0.183226	-0.0338971	0.438939	-0.00830934
Beam-2(C channel 100 X 10(1))	1	610.289	10.4593	1,817.59	111.939	98.2821	-3.65945
	2	43.468	129.298	1,646.49	92.8551	12.7634	-3.61881
	3	-67.1109	-275.919	915.263	-65.1978	-46.1411	-4.08685
	4	430.742	172.703	1,697.98	109.812	78.1145	-5.49818
	5	529.451	504.207	1,635.37	90.2951	69.1612	-3.91576
Beam-3(Angle iron 35 X 35 X 5(1)[1])	1	-15.4951	-0.947215	-0.155968	-0.357514	-0.512293	-0.0743896
	2	1.92144	-1.21578	0.440404	-0.000989849	0.0281502	-0.0094009
	3	-1.92144	1.21578	-0.440404	0.0513292	0.110817	0.0094009
	4	1.90646	-1.42381	0.524482	0.10369	0.224866	-0.00541984
Beam-4(Angle iron 35 X 35 X 5(1)[2])	1	-18.4284	-5.62489	-3.24603	-1.06317	1.89758	0.0568809
	2	-1.71954	-0.755441	-0.174202	0.0332027	-0.0782829	-0.0087817
	3	1.76437	0.774414	0.189333	0.0224455	-0.0803688	0.00509261
	4	1.71954	0.755441	0.174202	-0.0147894	-0.00156787	0.0087817
Beam-5(Trim/Extend1[5])	1	0	0	0	0	0	0
	2	-1.17485	0.622562	0.349395	0.0219093	-0.034512	-0.0115867
	3	1.17485	-0.622562	-0.349395	0.0108067	-0.0237821	0.0115867
	4	-1.17249	0.41453	0.434759	0.111542	-0.160177	-0.00481952
Beam-6(Cut-Extrude2)	1	5.27435	-0.712577	-0.0209532	-0.00332641	0.130175	-0.00101898
	2	1.83834	0.811257	-0.0790788	-0.0158629	-0.157471	0.00768663
	3	-0.592023	0.437629	-0.123897	0.021319	0.075602	0.00691245
	4	0.592019	-0.437634	0.123902	0.0194388	0.0683607	-0.00691256
	5	6.83147	0.637271	-0.0450416	-0.00716568	-0.107063	0.00414289

Beam-7(Cut-Extrude4)	1	-8.44054	-20.0835	-20.0835	-4.14063	4.55625	0.118327
	2	11.2517	-9.18737	-7.77465	-0.346451	-0.323228	-0.229709
	3	-21.8344	-25.1784	-9.22258	-0.0836989	2.7576	-0.20147
	4	-28.5685	29.7783	19.534	3.57401	-5.06004	-0.0315456
	5	-11.2517	9.18738	7.77464	3.23117	-3.08568	0.22971
Beam-8(Trim/Extend1[1])	1	0	0	0	0	0	0
	2	0	0	0	0	0	0
	3	-1.35849	0.182395	0.113356	0.0333069	0.0211939	0.00492276
	4	-1.31192	0.163422	0.105028	0.0092068	-0.00331738	0.0100057
Beam-9(Trim/Extend1[2])	1	55.9771	2.7569	27.1571	8.67484	0.626089	0.130641
	2	-0.293339	4.68342	-0.0188938	0.0157737	0.289119	-0.0372329
	3	0.293339	-4.68342	0.0188617	-0.00984805	1.18175	0.0372338
Beam-10(Trim/Extend1[4])	1	29.8163	8.50248	26.819	8.96941	-5.02841	-0.22245
	2	0.167006	-4.98991	0.188875	-0.0104017	-0.357889	0.0483838
	3	-0.167006	4.98991	-0.188842	-0.0489081	-1.20923	-0.0483847
Beam-11(Cut-Extrude2[5])	1	-78.0242	-49.335	107.519	-16.8984	-10.8345	-0.0343383
	2	78.0242	49.335	-107.519	-15.3572	-3.96604	0.0343383
Beam-12(Cut-Extrude2[13])	1	0	0	0	0	0	0
	2	0	0	0	0	0	0
Beam-13(C channel 100 X 10(1)[3])	1	-814.639	419.941	-11.7915	4.02815	19.707	0.544297
	2	1,379.29	-329.542	12.618	-0.680562	203.648	-0.23912
	3	1,315.96	343.384	-28.7371	-7.49306	93.5309	0.379008
	4	745.131	434.069	-40.8219	-9.66359	-20.0302	0.788177
	5	814.639	-419.941	11.7915	-0.0485074	122.023	-0.544297
Beam-15(Cut-Extrude5[6])	1	-1,585.21	142.72	-72.6915	-3.68448	29.1978	-0.0937033
	2	-1,542.45	142.046	15.3985	0.920542	113.401	0.0234903
	3	-1,899.23	-329.694	13.2622	0.274533	101.066	0.0666673

	4	-422.951	204.208	-4.80278	0.43166	6.10362	-0.0604663
	5	-2,724.19	-313.639	13.0265	0.6863	206.441	-0.145682
Beam-16(Cut-Extrude5[5])	1	-2,730.43	593.018	-80.3515	-7.20238	81.3294	-0.0511554
	2	-2,471.55	157.068	-70.2446	-7.80184	141.185	0.585912
	3	-985.347	-122.38	16.5412	2.54764	50.9443	-0.116449
	4	30.9277	84.6884	15.6313	-0.453459	3.39982	0.163705
	5	-3,579.15	-212.008	-61.7956	-2.95516	207.636	1.05569
Beam-17(C channel 100 X 10(1)[1])	1	-205.691	-94.2802	-29.9984	-5.58182	28.3638	0.245198
	2	-195.179	125.974	36.5757	4.72347	-14.0824	-0.227964
	3	86.7523	-83.6986	7.76718	-1.48916	-16.8331	-0.145503
	4	-193.342	-62.3136	18.4044	4.0169	3.96943	-0.23198
	5	-86.7523	83.6986	-7.76718	-1.13226	-11.4152	0.145503
Beam-18(Cut-Extrude5[1])	1	-1,724.47	643.811	-300.697	-27	13.4629	0.105358
	2	-131.532	42.3135	23.1317	-3.00602	20.1214	0.111198
	3	-2,410.94	897.582	101.676	9.12735	67.5677	0.0822392
	4	-306.547	-129.653	17.0629	2.12507	26.814	0.200513
	5	-3,825.33	-674.617	112.339	11.4285	182.28	-0.943816
Beam-19(Sb beam 100 X 8(1)[4])	1	-91.732	41.1581	5.4967	-0.798902	10.9256	0.0572294
	2	81.7748	0.835829	3.60738	0.835089	2.36867	-0.00770792
	3	11.5453	2.97026	10.9704	-0.72064	3.38599	0.0567217
	4	11.5453	2.97026	10.9704	1.85054	2.68984	0.0567217
	5	91.732	-41.1581	-5.49683	-1.05624	2.96523	-0.0572295
Beam-20(Cut-Extrude2[7])	1	457.102	71.2434	-95.791	-14.8268	-8.59889	0.0269527
	2	-457.102	-71.2434	95.791	-13.9105	-12.7741	-0.0269527

Beam-21(Cut-Extrude5[3])	1	-1,643.73	483.865	-19.4719	-1.30296	17.0568	-0.04151
	2	-2,107.24	113.092	-0.922514	-0.734555	100.433	-0.0414039
	3	-690.344	-165.349	-1.52512	-0.661604	97.425	0.00809732
	4	690.344	165.35	1.5251	-0.418784	19.7076	-0.00809698
	5	-1,966.59	-164.409	-0.702326	-0.547462	195.831	-0.0186602
Beam-22(Cut-Extrude2[11])	1	-636.178	-765.624	-61.4837	7.67462	-98.1648	-0.0129107
	2	636.178	765.624	61.4837	8.21142	-99.6556	0.0129107
Beam-23(C channel 100 X 10(1)[2])	1	1,793.9	-526.398	-23.7206	-9.15946	304.437	0.162825
	2	1,965.8	376.927	16.7528	-1.70431	168.057	-0.132411
	3	207.587	1,469.02	165.262	-32.8551	277.631	-1.24085
	4	-18.6824	1,388.26	114.559	18.6749	-207.169	-1.09792
	5	-207.587	-1,469.02	-165.262	-22.9209	218.163	1.24085
Beam-24(Cut-Extrude2[6])	1	10.9611	-41.3493	71.5166	-11.2418	-12.0358	0.0122016
	2	-10.9611	41.3493	-71.5166	-10.2132	-0.369031	-0.0122016
Beam-25(Cut-Extrude2[8])	1	465.321	998.041	47.8621	5.21483	-115.613	0.0400934
	2	-465.321	-998.041	-47.8621	7.1517	-142.259	-0.0400934
Beam-26(Cut-Extrude2[12])	1	3,248.82	1,595.34	-1,403.49	85.888	-187.375	0.240661
	2	-3,248.82	-1,595.34	1,403.49	147.538	452.709	-0.240661
Beam-27(Cut-Extrude5[2])	1	-2,214.33	482.338	147.214	13.4478	35.0668	-0.118986
	2	-2,598.88	868.421	-108.7	-9.69115	82.0635	0.0157373
	3	497.452	-732.219	34.1072	-1.99894	-29.3981	0.526476
	4	-112.177	-14.7163	-27.977	3.90514	9.6345	-0.0674716
	5	-3,297.44	-783.088	-6.40606	1.49994	200.648	-0.00121818
Beam-28(Cut-Extrude5[4])	1	-2,879.69	548.123	64.4018	7.85761	66.2896	0.272837

	2	-2,362.03	85.9303	46.923	4.96077	139.908	-0.352998
	3	-884.254	-97.0739	-25.093	-2.93212	44.5355	-0.0267802
	4	21.2837	71.9199	-3.65926	0.738673	2.58463	-0.0846383
	5	-3,532.16	-579.482	54.8345	3.21474	273.65	-1.43613
Beam-29(Cut-Extrude2[9])	1	-3,269.94	-1,418.99	-1,674.77	-131.686	455.534	0.0578428
	2	3,269.94	1,418.99	1,674.77	-62.3862	-291.102	-0.0578428

Beam Stresses

Beam Name	Joints	Axial(N/m^2)	Bending Dir1(N/m^2)	Bending Dir2(N/m^2)	Torsional (N/m^2)	Upper bound axial and bending(N/m^2)
Beam-1(Trim/Extend1[3])	1	-60,965.9	-1.52892e+06	-98,429.3	-294,170	1.68831e+06
	2	-1,554.89	-1,313.17	7,090.03	-55,839.3	9,958.09
	3	-1,554.89	42,458.1	-78,690.9	-55,841.3	122,704
Beam-2(C channel 100 X 10(1))	1	465,807	1.42328e+07	-2.45787e+06	-1.08921e+07	1.71565e+07
	2	33,177.2	1.18062e+07	-319,191	-1.07711e+07	1.21586e+07
	3	51,222.8	8.28971e+06	-1.15391e+06	1.21642e+07	9.49485e+06
	4	328,767	1.39622e+07	-1.95352e+06	-1.6365e+07	1.62445e+07
	5	404,107	1.14808e+07	-1.72961e+06	-1.1655e+07	1.36145e+07
Beam-3(Angle iron 35 X 35 X 5(1)[1])	1	-48,477.4	-65,452.3	343,361	-837,825	457,291
	2	6,011.35	-19,778.1	-32,543.6	-105,879	58,333.1
	3	6,011.35	15,301.1	91,510.3	-105,879	112,823
	4	5,964.5	-31,575.5	-185,998	-61,041.8	223,538
Beam-4(Angle iron 35 X 35 X 5(1)[2])	1	-57,654.5	53,328.3	-1.44489e+06	640,630	1.55588e+06
	2	5,379.69	14,272	-66,670.8	98,905.3	86,322.5
	3	5,519.95	-27,839.8	76,163	57,356.3	109,523
	4	5,379.69	-17,792.1	11,577.9	98,905.3	34,749.7
Beam-5(Trim/Extend1[5])	1	0	0	0	0	0
	2	-3,675.61	1,944.84	24,573.4	-130,509	30,193.8
	3	-3,675.61	3,520.48	-19,777.3	-130,509	26,973.4
	4	-3,668.22	20,192.1	107,517	-54,285.4	131,378
Beam-6(Cut-Extrude2)	1	60,961.7	162,519	-599,785	-89,748.4	823,266
	2	21,247.8	-886,324	1.0351e+06	677,011	1.94267e+06
	3	6,842.69	-900,789	710,253	-608,824	1.61789e+06

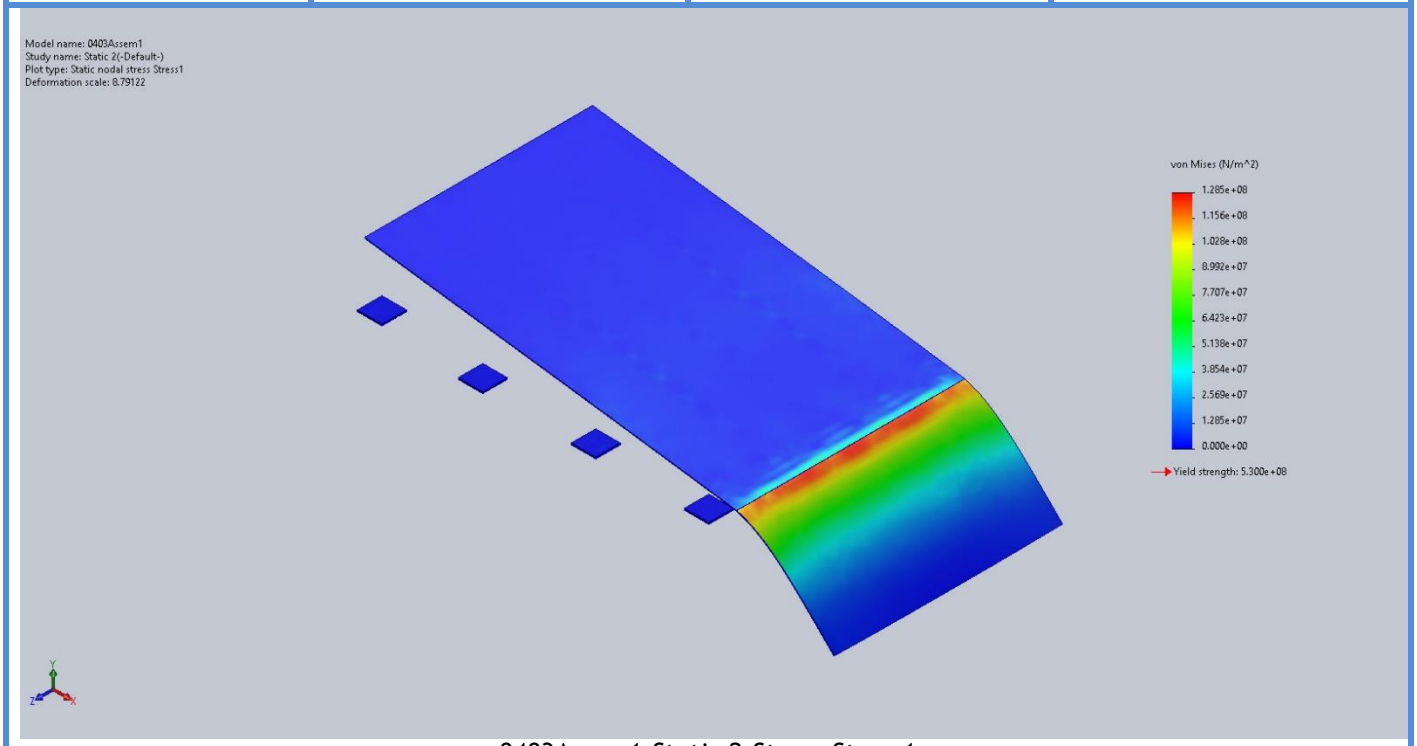
	4	6,842.64	820,119	-644,741	-608,834	1.4717e+06
	5	78,959.1	-477,073	647,416	364,891	1.20345e+06
Beam-7(Cut-Extrude4)	1	-11,797.5	-971,706	-127,404	936,160	1.11091e+06
	2	-15,726.7	179,167	-127,140	1.81737e+06	322,034
	3	-30,518.2	350,652	-523,981	-1.59395e+06	905,151
	4	-39,930.6	682,133	330,480	-249,576	1.05254e+06
	5	-15,726.7	823,548	7,516.22	1.81738e+06	846,791
Beam-8(Trim/Extend1[1])	1	0	0	0	0	0
	2	0	0	0	0	0
	3	-4,250.13	23,682.4	-1,933.21	55,467.5	29,865.8
	4	-4,104.42	12,627.8	9,859.74	112,740	26,591.9
Beam-9(Trim/Extend1[2])	1	65,372.9	1.69935e+06	-1.08258e+06	877,949	2.8473e+06
	2	342.576	-35,124	56,070	250,217	91,536.6
	3	342.576	129,604	-220,914	250,223	350,860
Beam-10(Trim/Extend1[4])	1	34,821	1.1257e+06	-53,064.1	-1.49493e+06	1.21358e+06
	2	-195.037	41,764.6	-68,392.1	-325,154	110,352
	3	-195.037	-143,699	232,613	-325,160	376,507
Beam-11(Cut-Extrude2[5])	1	71,579.8	3.50045e+06	-309,992	105,203	3.88202e+06
	2	71,579.8	-3.1812e+06	113,475	105,203	3.36626e+06
Beam-12(Cut-Extrude2[13])	1	0	0	0	0	0
	2	0	0	0	0	0
Beam-13(C channel 100 X 10(1)[3])	1	621,778	-512,168	492,839	-1.62006e+06	1.62679e+06
	2	1.05275e+06	-86,531.5	-5.09291e+06	-711,724	6.23219e+06
	3	1.00442e+06	-952,721	-2.33905e+06	1.12809e+06	4.29619e+06
	4	568,726	-1.2287e+06	500,922	2.34595e+06	2.29835e+06
	5	621,778	-6,167.57	-3.0516e+06	-1.62006e+06	3.67954e+06
Beam-15(Cut-Extrude5[6])	1	-1.45428e+06	-763,229	-835,398	-286,980	3.05291e+06
	2	-1.41505e+06	190,688	-3.24459e+06	71,942.4	4.85033e+06
	3	-1.74237e+06	56,868.7	-2.89168e+06	204,178	4.69091e+06
	4	388,018	-89,417.2	174,635	185,187	652,070
	5	-2.49918e+06	142,165	-5.90663e+06	-446,172	8.54798e+06
Beam-16(Cut-Extrude5[5])	1	-2.50491e+06	-1.49195e+06	-2.32697e+06	-156,671	6.32383e+06
	2	-2.26741e+06	-1.61613e+06	-4.03953e+06	1.79444e+06	7.92307e+06
	3	-903,962	527,736	-1.4576e+06	-356,642	2.8893e+06
	4	-28,373.2	93,932.7	97,274.5	-501,370	219,580

	5	-3.28353e+06	-612,154	-5.9408e+06	3.23319e+06	9.83649e+06
Beam-17(C channel 100 X 10(1)[1])	1	-156,995	-709,713	-709,332	729,814	1.57604e+06
	2	-148,972	600,576	352,177	-678,518	1.10173e+06
	3	-66,214.2	189,342	-420,968	433,078	676,525
	4	-147,569	510,737	-99,268.8	-690,472	757,575
	5	-66,214.2	-143,964	285,475	433,078	495,653
Beam-18(Cut-Extrude5[1])	1	-1.58204e+06	-5.59296e+06	-385,197	322,674	7.5602e+06
	2	120,668	622,688	575,706	-340,559	1.31906e+06
	3	-2.21181e+06	1.8907e+06	-1.93322e+06	251,869	6.03574e+06
	4	-281,227	440,203	-767,193	614,100	1.48862e+06
	5	-3.50938e+06	2.36738e+06	-5.21533e+06	-2.89058e+06	1.10921e+07
Beam-19(Sb beam 100 X 8(1)[4])	1	84,155.4	165,490	312,601	-175,295	562,246
	2	75,020.7	172,986	-67,771.6	-23,609.6	315,779
	3	10,591.7	-149,279	-96,878.8	173,740	256,749
	4	10,591.7	383,334	-76,960.6	173,740	470,886
	5	84,155.4	-218,798	-84,840.2	-175,296	387,794
Beam-20(Cut-Extrude2[7])	1	419,347	-3.07132e+06	246,028	82,575.7	3.7367e+06
	2	419,347	2.88152e+06	-365,489	82,575.7	3.66636e+06
Beam-21(Cut-Extrude5[3])	1	-1.50796e+06	-269,904	-488,024	-127,220	2.26589e+06
	2	-1.93319e+06	-152,161	-2.87355e+06	-126,895	4.9589e+06
	3	-633,326	-137,049	-2.78749e+06	24,816.7	3.55787e+06
	4	-633,326	86,749.9	563,868	24,815.6	1.28394e+06
	5	-1.80416e+06	-113,405	-5.60305e+06	-57,189.9	7.52061e+06
Beam-22(Cut-Extrude2[11])	1	583,633	-1.58978e+06	-2.80866e+06	39,554.8	4.98207e+06
	2	583,633	1.70097e+06	2.85131e+06	39,554.8	5.13592e+06
Beam-23(C channel 100 X 10(1)[2])	1	1.36921e+06	-1.1646e+06	-7.61347e+06	484,637	1.01473e+07
	2	1.50041e+06	-216,698	-4.20283e+06	-394,113	5.91993e+06
	3	-158,442	4.17742e+06	6.94308e+06	3.69332e+06	1.1279e+07
	4	-14,259.4	2.37446e+06	5.18096e+06	-3.26788e+06	7.56968e+06
	5	-158,442	-2.91432e+06	-5.45588e+06	3.69332e+06	8.52865e+06
Beam-24(Cut-Extrude2[6])	1	-10,055.7	2.32871e+06	-344,363	-37,382.3	2.68313e+06
	2	-10,055.7	-2.11563e+06	10,558.6	-37,382.3	2.13624e+06
Beam-25(Cut-Extrude2[8])	1	426,888	1.08024e+06	3.30787e+06	122,835	4.81499e+06
	2	426,888	-1.48145e+06	-4.07027e+06	122,835	5.97862e+06
Beam-26(Cut-Extrude2[12])	1	-2.98048e+06	-1.77915e+07	-5.36112e+06	-737,318	2.61331e+07
	2	-2.98048e+06	3.05621e+07	-1.29527e+07	-737,318	4.64953e+07
Beam-27(Cut-Extrude5[2])	1	-2.03144e+06	2.78567e+06	-1.00332e+06	-364,411	5.82042e+06
	2	-2.38423e+06	-2.00749e+06	-2.34797e+06	48,197.8	6.7397e+06
	3	456,365	-414,075	841,127	1.61241e+06	1.71157e+06
	4	102,912	-808,938	275,659	206,642	1.18751e+06

	5	-3.02509e+06	310,708	-5.74088e+06	-3,730.86	9.07667e+06
Beam-28(Cut-Extrude5[4])	1	-2.64185e+06	1.62768e+06	-1.89666e+06	835,603	6.16618e+06
	2	-2.16694e+06	1.02761e+06	-4.003e+06	-1.08111e+06	7.19755e+06
	3	-811,219	-607,380	-1.27423e+06	-82,018.2	2.69283e+06
	4	-19,525.7	-153,014	73,950.6	259,217	246,490
	5	-3.24043e+06	665,925	-7.82959e+06	-4.39835e+06	1.17359e+07
Beam-29(Cut-Extrude2[9])	1	-2.99986e+06	-2.72783e+07	-1.30336e+07	177,214	4.33117e+07
	2	-2.99986e+06	1.29231e+07	-8.32892e+06	177,214	2.42519e+07

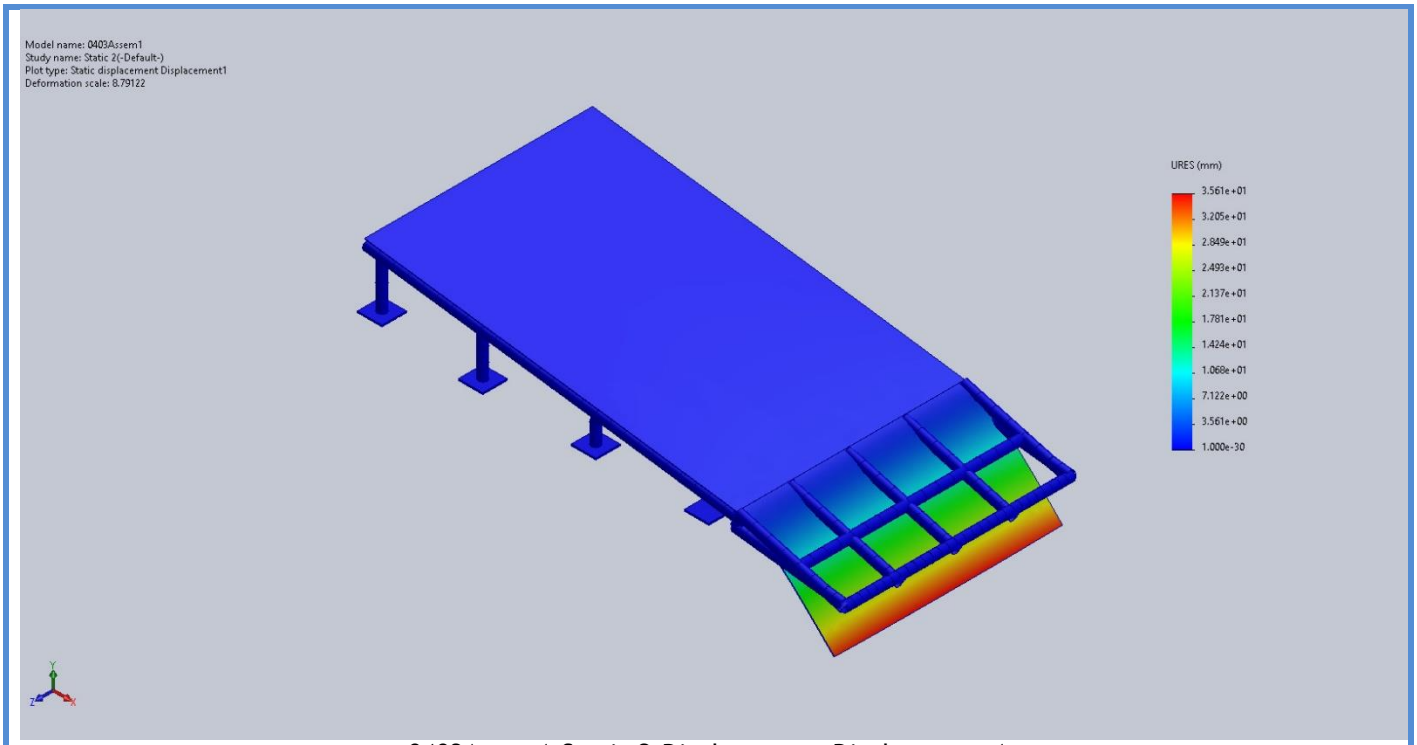
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	0.000e+00N/m^2 Node: 18254	1.285e+08N/m^2 Node: 14325



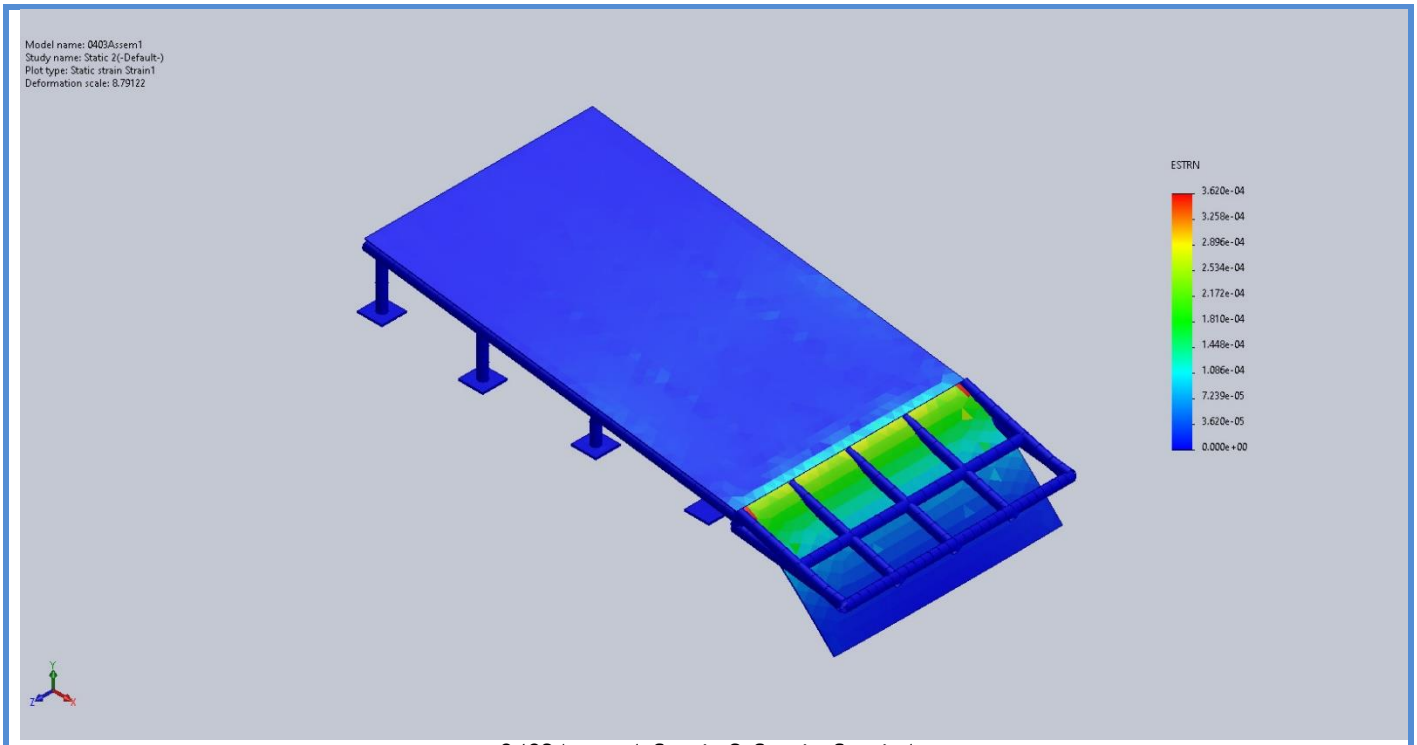
0403Assem1-Static 2-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 16939	3.561e+01mm Node: 12974



0403Assem1-Static 2-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	0.000e+00 Element: 8805	3.620e-04 Element: 8051



0403Assem1-Static 2-Strain-Strain1

Conclusion